King Fahd University of Petroleum & Minerals

Electrical Engineering Department

EE201-11: Fundamentals of Electric Circuits

First Semester 2002-2003 (011)

MAJOR I EXAM

Instructor: Dr. Saad Mohammad Al-Shahrani

Date: Friday 12 October 2001
Time: 6:00 PM to 7:30 PM
Location: Bldg 14-108

Name	Key Solution
ID#	

Question # 1	**35**	35
Question # 2	**35**	35
Question # 3	**30**	30
TOTAL	**100**	100

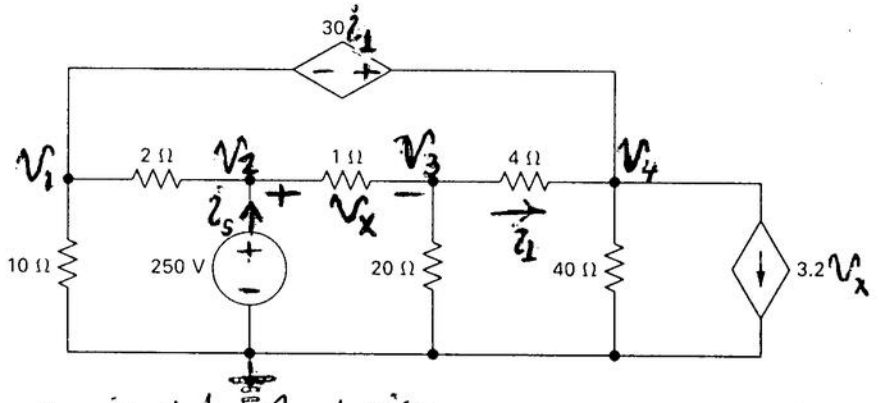
Question # 1:

For the circuit shown below, apply nodal method to find

a) v_1, v_2, v_3, and v_4 28

b) i_s 4

c) the power developed by the 250 V source. 3



a) Nodal Analysis:

at v_1: (Supernode) $\frac{v_1}{10} + \frac{v_1 - v_2}{2} + \frac{v_4}{40} + \frac{v_4 - v_3}{4} + 3.2(v$

$$24v_1 + 108v_2 - 138v_3 + 11v_4 = 0$$

at v_2: $v_2 = 250$

at v_3: $\frac{v_3 - v_2}{1} + \frac{v_3}{20} + \frac{v_3 - v_4}{4} = 0$

$$-20v_2 + 26v_3 - 5v_4 = 0$$

$v_4 - v_1 = 30\left(\frac{v_3 - v_4}{4}\right) \Rightarrow$ $4v_1 + 30v_3 - 34v_4 = 0$

$v_1 = 124.5$, $v_2 = 250$ V ; $v_3 = 235$, $v_4 =$

b) $i_s = \frac{v_2 - v_1}{2} + \frac{v_2 - v_3}{1} = 77.75$ A

c) $P_{250} = 250 \times i_s = 250 \times 77.75 = 19437.5$

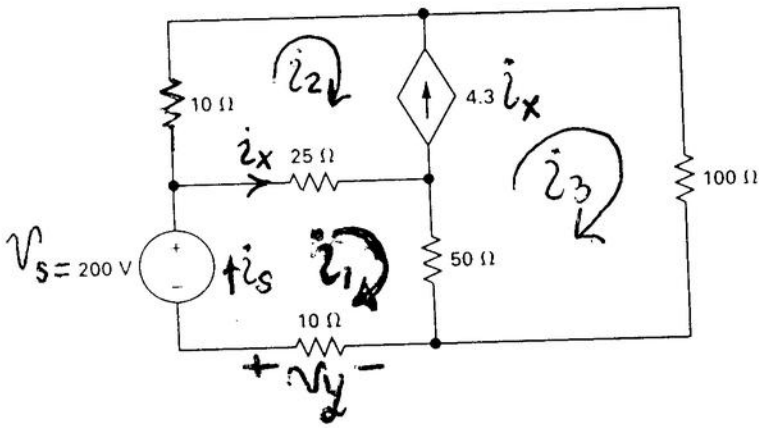
Question # 2:

For the circuit shown below

a) Use mesh analysis to find V_y.

b) F ind the resistance $R_{eq} = V_s/i_s$.

c) The power absorbed by the 100Ω resistor.



a) Loop1: KVL $\quad -200 + 25(i_1 - i_2) + 50(i_1 - i_3) + 10 i_1 = 0$

$$85 i_1 - 25 i_2 - 50 i_3 = 200 \quad (8)$$

Supermesh ② & ③

$$25(i_2 - i_1) + 10 i_2 + 100 i_3 + 50(i_3 - i_1) = 0$$

$$-75 i_1 + 35 i_2 + 150 i_3 = 0 \quad (8)$$

$$i_3 - i_2 = 4.3(i_1 - i_2)$$

$$4.3 i_1 - 3.3 i_2 - i_3 = 0 \quad (8)$$

$i_1 = 4.6\ A$; $i_2 = 5.7\ A$; $i_3 = 0.97\ A$

$V_y = -10 i_1 = -10(4.6) = -46\ V$ (4)

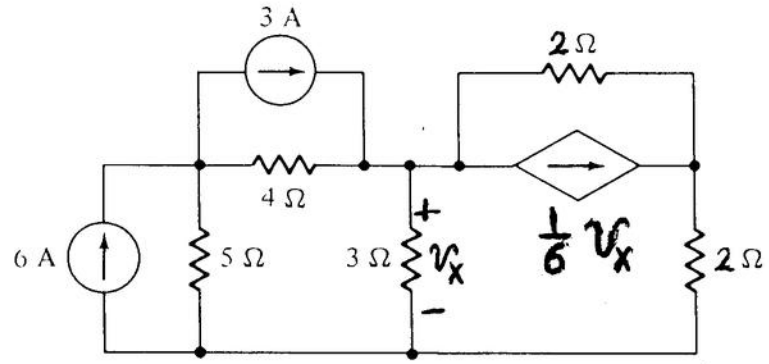
b) $i_s = i_1 = 4.6\,A$

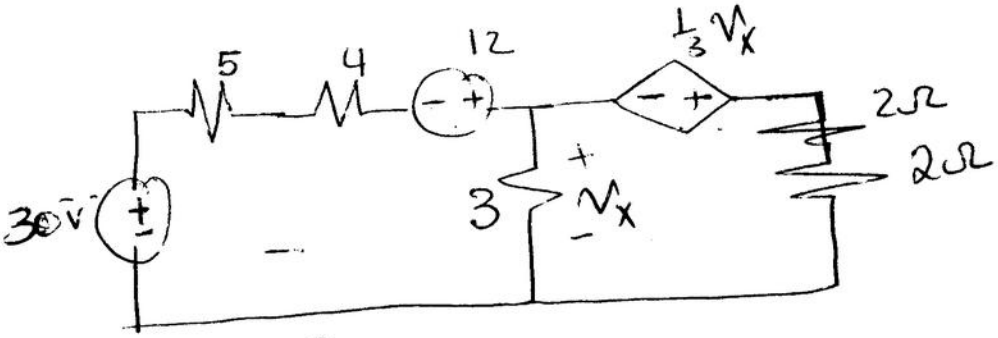
$R_{eq} = \frac{V_s}{i_s} = \frac{200}{4.6} = 43.5\,\Omega$ (3)

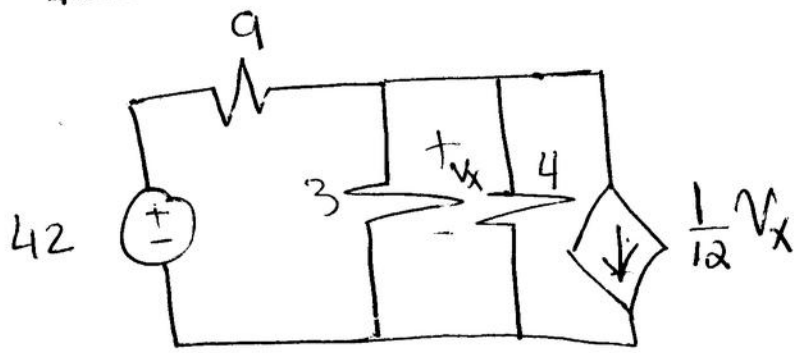
c) $P_{100} = i_3^2(100) = (0.97)^2(100) = 94.00$

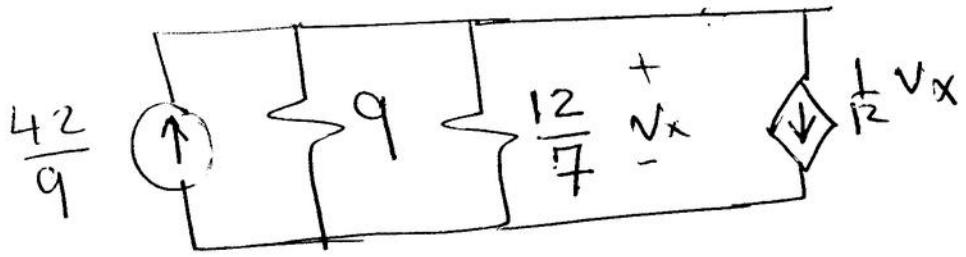
Question # 3:

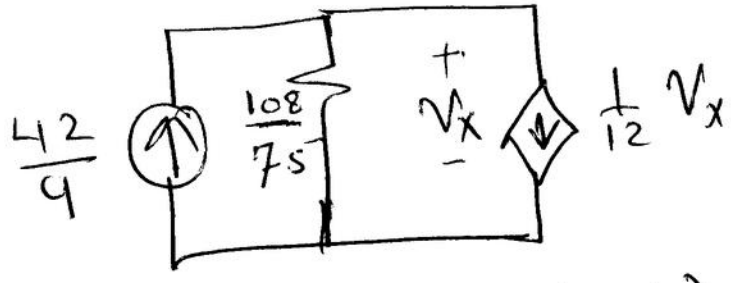
For the circuit shown below, use the source transformations (at least 4 times) to determine V_x.











$$V_x = \frac{108}{75}\left(\frac{42}{9} - \frac{1}{12}V_x\right) \Rightarrow \boxed{V_x = 6\ V}$$